

Question	Answer	Marks
3(a)(i)	number of particles per unit amount of substance	B1
3(a)(ii)	$N_A = R / k$	B1
3(b)(i)	X pressure and Y pressure <u>both</u> = NkT / V	B1
	X amount = N / N_A and Y amount = $2N / N_A$	B1
	X mean-square speed = $3kT / m$ and Y mean-square speed = $3kT / 2m$	B1
	X internal energy = $3NkT / 2$ and Y internal energy = $3NkT$	B1
3(b)(ii)	line passing through the origin and not returning to either axis	B1
	curve with positive decreasing gradient	B1

9702/41

Cambridge International AS & A Level – Mark Scheme

October/November 2024

PUBLISHED

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